

SEMICONDUCTOR MISSION AND ELECTRONICS MANUFACTURING

DESIGN-TO-DEVICE RAIL → SEMICONDUCTOR BUSINESS-MODEL MATRIX → FRONT-END VERSUS BACK-END → NODE AND LITHOGRAPHY LADDER → ADVANCED INTEGRATION MAP → MATERIALS TAXONOMY

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • active generation g1 • explicit approval of this exact generation is required • all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

00

STAGE 00

DESIGN-TO-DEVICE RAIL

DESIGN-TO-DEVICE RAIL

ARCHITECTURE, VERIFICATION AND TAPE-OUT

TRANSISTOR AND INTERCONNECT LAYERS ON WAFER

WAFER BECOMES INDIVIDUAL DIES

ASSEMBLE, TEST, MARK AND PACKAGE

DOWNSTREAM ELECTRONICS INTEGRATION

DESIGN / IP / EDA → architecture, verification and tape-out

FOUNDRY / FAB → transistor and interconnect layers on wafer

DICING → wafer becomes individual dies

ATMP / OSAT → assemble, test, mark and package

BOARD / MODULE / DEVICE → downstream electronics integration

STARTING CONDITION / INPUT

- DESIGN / IP / EDA → architecture, verification and tape-out
- FOUNDRY / FAB → transistor and interconnect layers on wafer

SCIENTIFIC OR ENGINEERING MECHANISM

- DICING → wafer becomes individual dies
- ATMP / OSAT → assemble, test, mark and package

OUTPUT / FAILURE BOUNDARY

- BOARD / MODULE / DEVICE → downstream electronics integration
- DESIGN / IP / EDA → architecture, verification and tape-out

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: BOARD / MODULE / DEVICE → downstream electronics integration.

01

STAGE 01

SEMICONDUCTOR BUSINESS-MODEL MATRIX

SEMICONDUCTOR BUSINESS-MODEL MATRIX

PHYSICAL WAFER-MANUFACTURING PLANT

DESIGNS AND OUTSOURCES WAFER MANUFACTURE

MANUFACTURES TO CUSTOMERS' DESIGNS

DESIGNS AND FABRICATES ITS OWN PRODUCTS

PLANT TYPE AND FIRM MODEL ANSWER DIFFERENT QUESTIONS

SYSTEM / DEVICE / INSTITUTION

FAB → physical wafer-manufacturing plant

FABLESS → designs and outsources wafer manufacture

DECISIVE TECHNICAL DISTINCTION

FOUNDRY → manufactures to customers' designs

IDM → designs and fabricates its own products

STATUS / UNIT / EVIDENCE LIMIT

RULE → plant type and firm model answer different questions

FAB → physical wafer-manufacturing plant

SYSTEM / DEVICE / INSTITUTION

- FAB → physical wafer-manufacturing plant
- FABLESS → designs and outsources wafer manufacture

DECISIVE TECHNICAL DISTINCTION

- FOUNDRY → manufactures to customers' designs
- IDM → designs and fabricates its own products

STATUS / UNIT / EVIDENCE LIMIT

- RULE → plant type and firm model answer different questions
- FAB → physical wafer-manufacturing plant

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → plant type and firm model answer different questions.

02

STAGE 02

FRONT-END VERSUS BACK-END

FRONT-END VERSUS BACK-END

FRONT END

DEPOSITION, LITHOGRAPHY, ETCHING, DOPING, METALLISATION

PROCESSED WAFER CONTAINING DIES

BACK END

BONDING, ENCAPSULATION, TESTING, MARKING, PACKAGING

PROCESS DESCRIPTION; OSAT

OUTSOURCED SERVICE MODEL

SYSTEM / DEVICE / INSTITUTION

FRONT END → deposition, lithography, etching, doping, metallisation

OUTPUT → processed wafer containing dies

DECISIVE TECHNICAL DISTINCTION

BACK END → bonding, encapsulation, testing, marking, packaging

ATMP → process description; OSAT → outsourced service model

STATUS / UNIT / EVIDENCE LIMIT

TRAP → back-end capacity is not front-end fabrication

FRONT END → deposition, lithography, etching, doping, metallisation

SYSTEM / DEVICE / INSTITUTION

- FRONT END → deposition, lithography, etching, doping, metallisation
- OUTPUT → processed wafer containing dies

DECISIVE TECHNICAL DISTINCTION

- BACK END → bonding, encapsulation, testing, marking, packaging
- ATMP → process description; OSAT → outsourced service model

STATUS / UNIT / EVIDENCE LIMIT

- TRAP → back-end capacity is not front-end fabrication
- FRONT END → deposition, lithography, etching, doping, metallisation

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRAP → back-end capacity is not front-end fabrication.

03

STAGE 03

NODE AND LITHOGRAPHY LADDER

NODE AND LITHOGRAPHY LADDER

BROAD AUTOMOTIVE, INDUSTRIAL AND POWER USES

MATURE AND MID-RANGE LITHOGRAPHY ROUTE

LEADING EDGE

RIISING COMPLEXITY AND CAPITAL INTENSITY

CONCENTRATED EQUIPMENT AND EXPORT-CONTROL CHOKEPOINT

VERIFY EVERY PROJECT-SPECIFIC NODE CLAIM OFFICIALLY

MATURE / MID-RANGE → broad automotive, industrial and power uses

DUV → mature and mid-range lithography route

LEADING EDGE → rising complexity and capital intensity

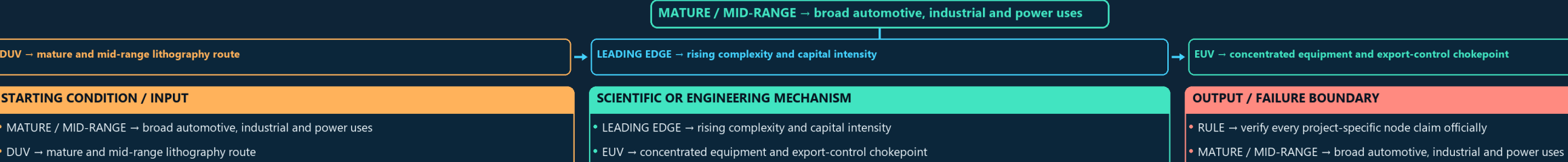
EUV → concentrated equipment and export-control chokepoint

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRAP → back-end capacity is not front-end fabrication.

03

STAGE 03 **NODE AND LITHOGRAPHY LADDER**

NODE AND LITHOGRAPHY LADDER **BROAD AUTOMOTIVE, INDUSTRIAL AND POWER USES** **MATURE AND MID-RANGE LITHOGRAPHY ROUTE** **LEADING EDGE** **RISING COMPLEXITY AND CAPITAL INTENSITY** **CONCENTRATED EQUIPMENT AND EXPORT-CONTROL CHOKEPOINT**
VERIFY EVERY PROJECT-SPECIFIC NODE CLAIM OFFICIALLY

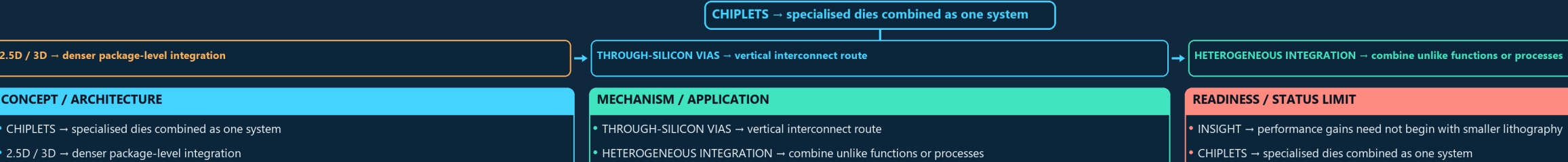


ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → verify every project-specific node claim officially.

04

STAGE 04 **ADVANCED INTEGRATION MAP**

ADVANCED INTEGRATION MAP **SPECIALISED DIES COMBINED AS ONE SYSTEM** **DENSER PACKAGE-LEVEL INTEGRATION** **THROUGH-SILICON VIAS** **VERTICAL INTERCONNECT ROUTE** **HETEROGENEOUS INTEGRATION** **COMBINE UNLIKE FUNCTIONS OR PROCESSES**
PERFORMANCE GAINS NEED NOT BEGIN WITH SMALLER LITHOGRAPHY

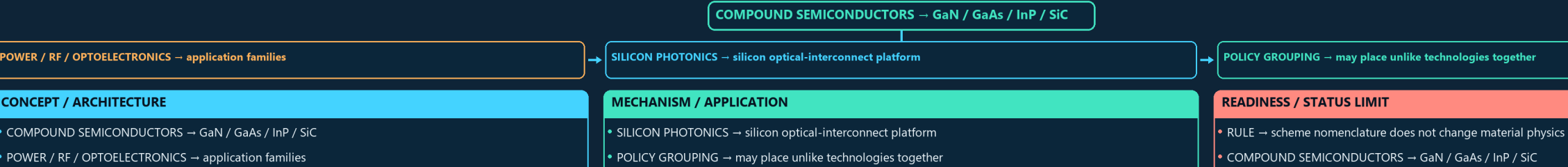


ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: INSIGHT → performance gains need not begin with smaller lithography.

05

STAGE 05 **MATERIALS TAXONOMY**

MATERIALS TAXONOMY **COMPOUND SEMICONDUCTORS** **APPLICATION FAMILIES** **SILICON PHOTONICS** **SILICON OPTICAL-INTERCONNECT PLATFORM** **POLICY GROUPING** **MAY PLACE UNLIKE TECHNOLOGIES TOGETHER** **SCHEME NOMENCLATURE DOES NOT CHANGE MATERIAL PHYSICS**

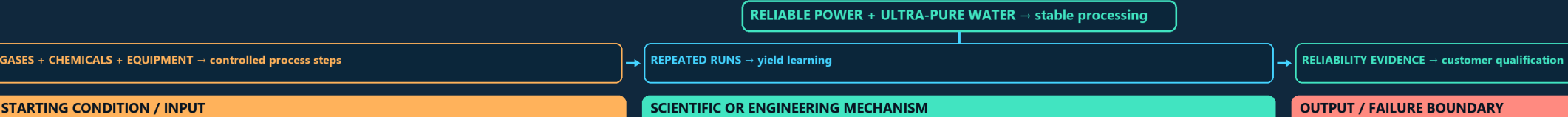


ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → scheme nomenclature does not change material physics.

06

STAGE 06 **FAB VIABILITY CHAIN**

FAB VIABILITY CHAIN **RELIABLE POWER + ULTRA-PURE WATER** **STABLE PROCESSING** **GASES + CHEMICALS + EQUIPMENT** **CONTROLLED PROCESS STEPS** **REPEATED RUNS** **YIELD LEARNING** **RELIABILITY EVIDENCE**



ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → scheme nomenclature does not change material physics.

06

STAGE 06 FAB VIABILITY CHAIN

FAB VIABILITY CHAIN

RELIABLE POWER + ULTRA-PURE WATER

STABLE PROCESSING

GASES + CHEMICALS + EQUIPMENT

CONTROLLED PROCESS STEPS

REPEATED RUNS

YIELD LEARNING

RELIABILITY EVIDENCE

RELIABLE POWER + ULTRA-PURE WATER → stable processing

GASES + CHEMICALS + EQUIPMENT → controlled process steps

→ REPEATED RUNS → yield learning

→ RELIABILITY EVIDENCE → customer qualification

STARTING CONDITION / INPUT

- RELIABLE POWER + ULTRA-PURE WATER → stable processing
- GASES + CHEMICALS + EQUIPMENT → controlled process steps

SCIENTIFIC OR ENGINEERING MECHANISM

- REPEATED RUNS → yield learning
- RELIABILITY EVIDENCE → customer qualification

OUTPUT / FAILURE BOUNDARY

- OUTCOME → inauguration alone is not competitive production
- RELIABLE POWER + ULTRA-PURE WATER → stable processing

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: OUTCOME → inauguration alone is not competitive production.

07

STAGE 07 INSTITUTION AND SCHEME MAP

INSTITUTION AND SCHEME MAP

PARENT MINISTRY

IMPLEMENTING AGENCY

SEMICON INDIA

SCHEME PACKAGE

DESIGN-FOCUSED CHANNEL

SEPARATE COMPLEMENTARY ELECTRONICS INCENTIVES

MeitY → parent ministry

ISM → implementing agency

→ SEMICON INDIA → scheme package

→ DLI → design-focused channel

CONCEPT / ARCHITECTURE

- MeitY → parent ministry
- ISM → implementing agency

MECHANISM / APPLICATION

- SEMICON INDIA → scheme package
- DLI → design-focused channel

READINESS / STATUS LIMIT

- SPECS / PLI → separate complementary electronics incentives
- MeitY → parent ministry

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SPECS / PLI → separate complementary electronics incentives.

08

STAGE 08 SEMICON 1.0 TO 2.0

SEMICON 1.0 TO 2.0

SEMICON INDIA 1.0

RS 76,000 CRORE UMBRELLA

FOUR CHANNELS

FABS, DISPLAYS, COMPOUND/SENSOR/ATMP, DLI

15 JUL 2026

SEMICON 2.0 CABINET APPROVAL

SEMICON 2.0

SEMICON INDIA 1.0 → Rs 76,000 crore umbrella

→ FOUR CHANNELS → fabs, displays, compound/sensor/ATMP, DLI

→ 15 JUL 2026 → Semicon 2.0 Cabinet approval

→ SEMICON 2.0 → Rs 1,27,500 crore approved outlay

→ SIX PILLARS → design, tools/materials, fabs, back end, R&D, talent

CONCEPT / ARCHITECTURE

- SEMICON INDIA 1.0 → Rs 76,000 crore umbrella
- FOUR CHANNELS → fabs, displays, compound/sensor/ATMP, DLI

MECHANISM / APPLICATION

- 15 JUL 2026 → Semicon 2.0 Cabinet approval
- SEMICON 2.0 → Rs 1,27,500 crore approved outlay

READINESS / STATUS LIMIT

- SIX PILLARS → design, tools/materials, fabs, back end, R&D, talent
- SEMICON INDIA 1.0 → Rs 76,000 crore umbrella

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SIX PILLARS → design, tools/materials, fabs, back end, R&D, talent.

09

STAGE 09 FACILITY EVIDENCE MAP

FACILITY EVIDENCE MAP

FAB; LATER NAMED COMPOUND-FAB PLUS ATMP APPROVAL

OFFICIALLY IDENTIFIED OSAT

ATMP LOCATION

OFFICIALLY IDENTIFIED BACK-END CLUSTER AND INAUGURATIONS

SUCHI SEMICON OSAT APPROVAL

LOCATION MUST TRAVEL WITH PROJECT AND STATUS VERB

DHOLERA → fab; later named compound-fab plus ATMP approval

MORIGAON → officially identified OSAT / ATMP location

→ SANAND → officially identified back-end cluster and inaugurations

→ SURAT → Suchi Semicon OSAT approval

CONCEPT / ARCHITECTURE

- DHOLERA → fab; later named compound-fab plus ATMP approval
- MORIGAON → officially identified OSAT / ATMP location

MECHANISM / APPLICATION

- SANAND → officially identified back-end cluster and inaugurations
- SURAT → Suchi Semicon OSAT approval

READINESS / STATUS LIMIT

- RULE → location must travel with project and status verb
- DHOLERA → fab; later named compound-fab plus ATMP approval

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → location must travel with project and status verb.

